Smart Electricity Billing for Multiple Apartments

Python Programming – Activity-Based Learning (ABL) Assignment

1. Problem Statement

Calculate apartment-wise electricity bills using progressive tariffs, fixed charges, and rebates; identify billing discrepancies and rank apartments by consumption.

2. Objectives

- Calculate consumption from meter readings.
- Apply progressive tariff slabs.
- Add fixed charges and deduct rebates.
- Compare calculated and recorded bills.
- Rank apartments by units consumed.

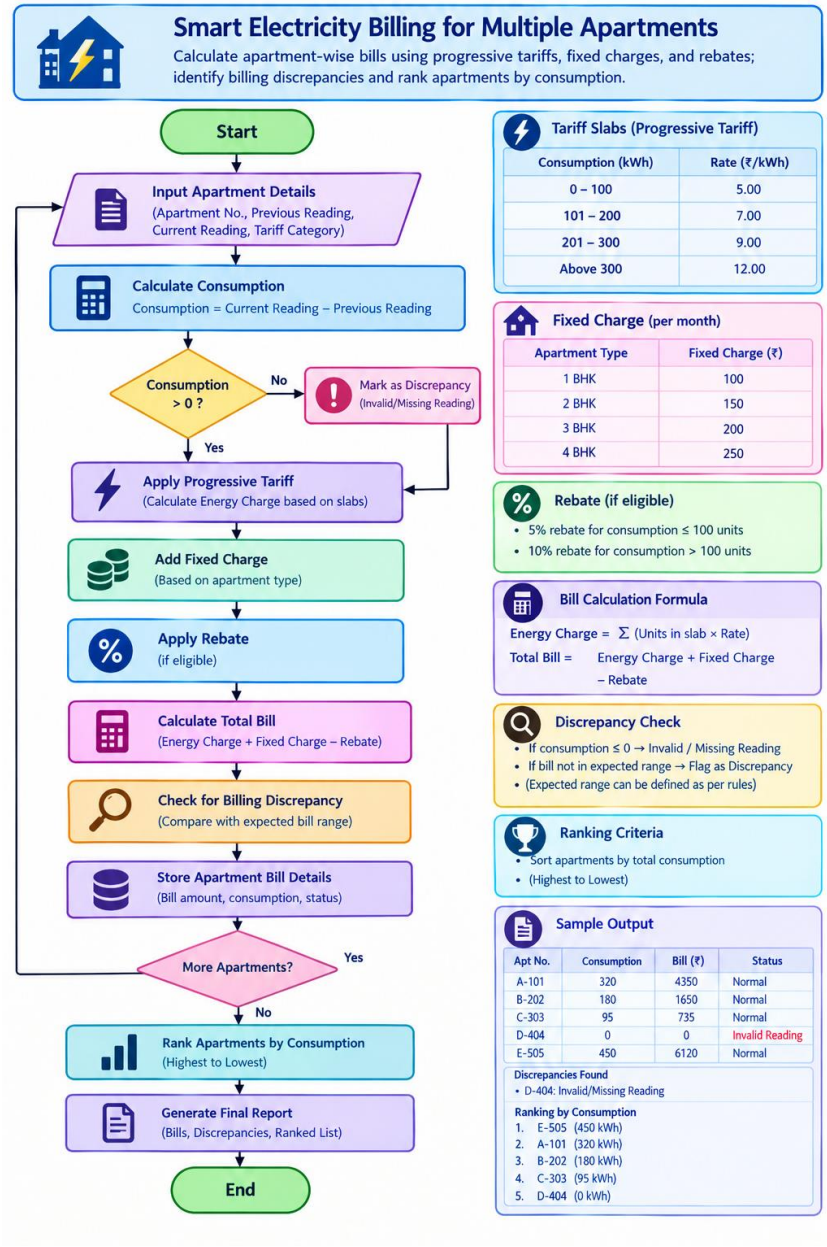
3. Assumptions and Tariff Policy

The first 100 units cost Rs. 2 per unit, the next 100 units cost Rs. 3 per unit, and units above 200 cost Rs. 5 per unit. A fixed charge of Rs. 100 is added. The specified rebate is deducted from the bill, with a minimum bill of zero. Recorded bills are compared against calculated bills to flag discrepancies.

4. Algorithm

1. Store apartment ID, previous/current meter readings, rebate, and recorded bill.
2. Calculate units = current reading - previous reading.
3. Calculate energy charge using progressive tariff slabs.
4. Add fixed charge and subtract rebate.
5. Compare calculated bill with recorded bill and flag mismatches.
6. Sort apartments by consumption in descending order.
7. Print apartment-wise bills, rankings, and billing summary.

5. Flowchart



6. Pseudocode

BEGIN

DECLARE apartments as empty list

INPUT number_of_apartments

```
FOR i = 1 TO number_of_apartments DO

   INPUT apartment_number
   INPUT apartment_type
   INPUT previous_reading
   INPUT current_reading

   IF previous_reading < 0 OR
     current_reading < previous_reading THEN

      status = "Invalid Reading"
      consumption = 0
      total_bill = 0

      DISPLAY "Billing Discrepancy Detected"

   ELSE

      consumption = current_reading - previous_reading

      // Calculate Progressive Energy Charges
      energy_charge = 0

      IF consumption <= 100 THEN
         energy_charge = consumption * 5
```

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ELSE IF consumption <= 200 THEN
   energy_charge = (100 * 5) +
                   ((consumption - 100) * 7)

ELSE IF consumption <= 300 THEN
   energy_charge = (100 * 5) +
                   (100 * 7) +
                   ((consumption - 200) * 9)

ELSE
   energy_charge = (100 * 5) +
                   (100 * 7) +
                   (100 * 9) +
                   ((consumption - 300) * 12)
END IF

// Fixed Charges
IF apartment_type = "1 BHK" THEN
   fixed_charge = 100
ELSE IF apartment_type = "2 BHK" THEN
   fixed_charge = 150
ELSE IF apartment_type = "3 BHK" THEN
   fixed_charge = 200
ELSE IF apartment_type = "4 BHK" THEN
   fixed_charge = 250
```

```
    ELSE
        fixed_charge = 0
    END IF

    // Calculate Rebate
    IF consumption <= 100 THEN
        rebate = energy_charge * 5 / 100
    ELSE
        rebate = energy_charge * 10 / 100
    END IF

    // Calculate Final Bill
    total_bill = energy_charge +
                 fixed_charge - rebate

    status = "Normal"

    // Check Billing Discrepancy
    IF total_bill < 0 THEN
        status = "Billing Discrepancy"
    END IF

END IF

STORE apartment_number,
      consumption,
```

```
        total_bill,
        status IN apartments

END FOR

// Rank Valid Apartments by Consumption
SORT valid apartments BY consumption
    IN DESCENDING ORDER

// Generate Final Report
DISPLAY "SMART ELECTRICITY BILLING REPORT"

FOR each apartment IN apartments DO
   DISPLAY apartment_number
   DISPLAY consumption
   DISPLAY total_bill
   DISPLAY status
END FOR

DISPLAY "APARTMENTS RANKED BY CONSUMPTION"

FOR each valid apartment IN sorted apartments DO
   DISPLAY apartment_number
   DISPLAY consumption
END FOR
```

```
  DISPLAY "BILLING DISCREPANCIES"

  FOR each apartment IN apartments DO
    IF status = "Invalid Reading" OR
      status = "Billing Discrepancy" THEN
       DISPLAY apartment_number
       DISPLAY status
    END IF
  END FOR

  DISPLAY "Report Generated Successfully"

END
```

7. Python Source Code

```
def calculate_energy_charge(units): """Calculate electricity charges using progressive tariff slabs.""" if units <= 100: return units * 5 elif units <= 200: return (100 * 5) + ((units - 100) * 7) elif units <= 300: return (100 * 5) + (100 * 7) + ((units - 200) * 9) else: return (100 * 5) + (100 * 7) + (100 * 9) + ((units - 300) * 12)

def get_fixed_charge(apartment_type): charges = { "1 BHK": 100, "2 BHK": 150, "3 BHK": 200, "4 BHK": 250 } return charges.get(apartment_type.upper(), None)

def calculate_rebate(units, energy_charge): if units <= 100: return energy_charge * 0.05 else: return energy_charge * 0.10

def process_apartment(): apartment_number = input("Enter apartment number: ") apartment_type = input("Enter apartment type (1 BHK/2 BHK/3 BHK/4 BHK): ").strip()


    previous_reading = float(input("Enter previous meter reading: "))
    current_reading = float(input("Enter current meter reading: "))
except ValueError:
```

```
try:
    print("Invalid input: Meter readings must be numbers.")
    return None

# Validate readings
if previous_reading < 0 or current_reading < previous_reading:
    print("Billing Discrepancy: Invalid or missing meter reading.")
    return {
        "number": apartment_number,
        "type": apartment_type,
        "consumption": 0,
        "energy_charge": 0,
        "fixed_charge": 0,
        "rebate": 0,
        "bill": 0,
        "status": "Invalid Reading"
    }

fixed_charge = get_fixed_charge(apartment_type)

if fixed_charge is None:
    print("Invalid apartment type.")
    return None

consumption = current_reading - previous_reading

energy_charge = calculate_energy_charge(consumption)
rebate = calculate_rebate(consumption, energy_charge)

total_bill = energy_charge + fixed_charge - rebate

# Check for unusually high bills
status = "Normal"
if total_bill < 0:
    status = "Billing Discrepancy"

return {
    "number": apartment_number,
    "type": apartment_type,
```

```
    "consumption": consumption,
    "energy_charge": energy_charge,
    "fixed_charge": fixed_charge,
    "rebate": rebate,
    "bill": total_bill,
    "status": status
}
```

def main(): print("=" * 75) print(" SMART ELECTRICITY BILLING FOR MULTIPLE APARTMENTS") print("=" * 75)

```
apartments = []

try:
    n = int(input("Enter number of apartments: "))
            print("Number of apartments must be positive.")
        return
except ValueError:
    print("Please enter a valid whole number.")
    return

for i in range(n):
    print(f"\nEnter details for Apartment {i + 1}")
    apartment = process_apartment()

    if apartment is not None:
        apartments.append(apartment)

# Display apartment-wise bills
print("\n" + "=" * 100)
print("                    APARTMENT-WISE BILLING REPORT")
print("=" * 100)

print(
    f"{'Apartment':<12}{'Type':<10}{'Units':>10}"
    f"{'Energy Rs.':>13}{'Fixed Rs.':>12}"
    f"{'Rebate Rs.':>13}{'Bill Rs.':>12}{'Status':>18}"
)
print("-" * 100)
```

```
for apt in apartments:
    print(
        f"{apt['number']:<12}{apt['type']:<10}"
        f"{apt['consumption']:>10.2f}"
        f"{apt['energy_charge']:>13.2f}"
        f"{apt['fixed_charge']:>12.2f}"
        f"{apt['rebate']:>13.2f}"
        f"{apt['bill']:>12.2f}"
        f"{apt['status']:>18}"
    )

# Identify discrepancies
print("\n" + "=" * 75)
print("                 BILLING DISCREPANCIES")
print("=" * 75)

discrepancies = [
    apt for apt in apartments
    if apt["status"] != "Normal"
]

if discrepancies:
    for apt in discrepancies:
        print(
            f"Apartment {apt['number']}: {apt['status']}"
        )
else:
    print("No billing discrepancies found.")

# Rank valid apartments by consumption
valid_apartments = [
    apt for apt in apartments
    if apt["status"] == "Normal"
]

ranked = sorted(
    valid_apartments,
    key=lambda apt: apt["consumption"],
```

```
    reverse=True
)

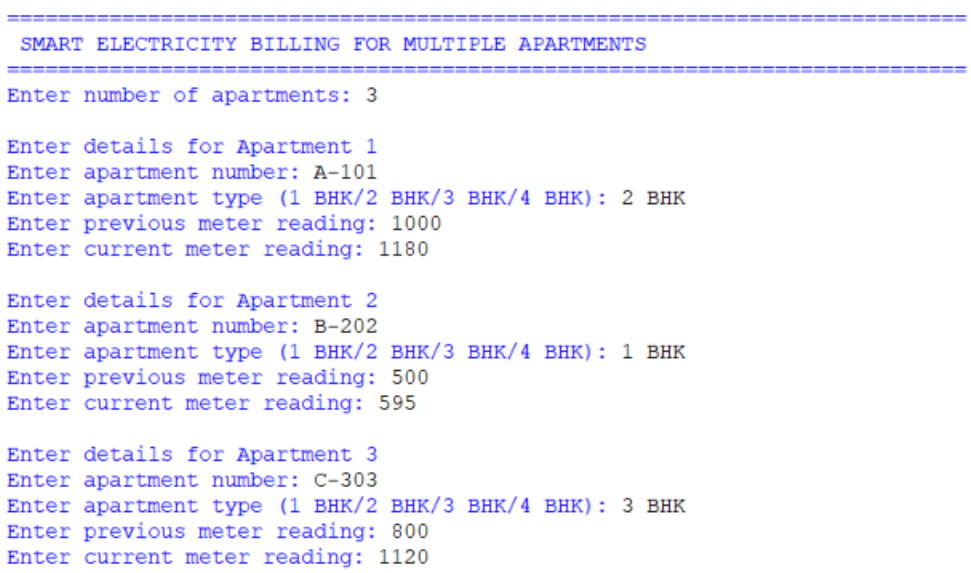
print("\n" + "=" * 75)
print("              CONSUMPTION RANKING (HIGHEST FIRST)")
print("=" * 75)

if ranked:
    for rank, apt in enumerate(ranked, start=1):
        print(
            f"{rank}. Apartment {apt['number']} - "
            f"{apt['consumption']:.2f} units - "
            f"Bill: Rs. {apt['bill']:.2f}"
        )
else:
    print("No valid apartments available for ranking.")

print("\n" + "=" * 75)
print("                   REPORT COMPLETED")
print("=" * 75)
```

if **name** == "**main**": main()

8. Output Screenshot



```
================================================================================
============
                    APARTMENT-WISE BILLING REPORT
================================================================================
============
Apartment    Type           Units    Energy Rs.    Fixed Rs.    Rebate Rs.    Bill Rs.
     Status
--------------------------------------------------------------------------------
------------
A-101        2 BHK         180.00      1060.00       150.00        106.00     1104.00
     Normal
B-202        1 BHK          95.00       475.00       100.00         23.75      551.25
     Normal
C-303        3 BHK         320.00      2340.00       200.00        234.00     2306.00
     Normal

================================================================================
                  BILLING DISCREPANCIES
================================================================================
No billing discrepancies found.

================================================================================
              CONSUMPTION RANKING (HIGHEST FIRST)
================================================================================
1. Apartment C-303 - 320.00 units - Bill: Rs. 2306.00
2. Apartment A-101 - 180.00 units - Bill: Rs. 1104.00
3. Apartment B-202 - 95.00 units - Bill: Rs. 551.25

================================================================================
                   REPORT COMPLETED
================================================================================
```

9. Test Cases

1. **TC01 – Valid Meter Readings:** Enter valid previous and current meter readings. Verify that electricity consumption is calculated correctly.

2. **TC02 – Progressive Tariff (0–100 Units):** Enter consumption of 80 units. Verify that the energy charge is calculated at ₹5 per unit.

3. **TC03 – Progressive Tariff (101–200 Units):** Enter consumption of 180 units. Verify that the energy charge is ₹1,060.

4. **TC04 – Progressive Tariff (201–300 Units):** Enter consumption of 250 units. Verify that the energy charge is ₹1,850.

5. **TC05 – Progressive Tariff (Above 300 Units):** Enter consumption of 350 units. Verify that the energy charge is ₹2,600.

6. **TC06 – Fixed Charges:** Select 1 BHK, 2 BHK, 3 BHK, or 4 BHK. Verify that the fixed charge is ₹100, ₹150, ₹200, or ₹250, respectively.

7. **TC07 – Rebate Calculation:** Enter consumption of 80 units. Verify that a 5% rebate on the energy charge is applied.

8. **TC08 – High Consumption Rebate:** Enter consumption of 180 units. Verify that a 10% rebate on the energy charge is applied.

9. **TC09 – Invalid Meter Reading:** Enter a current reading lower than the previous reading. Verify that an invalid reading discrepancy is reported.

10. **TC10 – Negative Meter Reading:** Enter a negative meter reading. Verify that the system identifies it as invalid.

11. **TC11 – Invalid Apartment Type:** Enter an unsupported apartment type. Verify that an error message is displayed and no bill is calculated for that entry.

12. **TC12 – Apartment-wise Bill:** Enter valid details for multiple apartments. Verify that energy charges, fixed charges, rebates, and final bills are displayed correctly.

13. **TC13 – Billing Discrepancy Report:** Include an apartment with invalid readings. Verify that it appears in the discrepancy report.

14. **TC14 – Consumption Ranking:** Enter different consumption values for multiple apartments. Verify that valid apartments are ranked from highest to lowest consumption.

15. **TC15 – Invalid Number of Apartments:** Enter zero, a negative number, or non-numeric input. Verify that the program handles the invalid input appropriately.

16. **TC16 – Final Report:** Complete the billing process and verify that the apartment-wise billing report, discrepancy list, and consumption ranking are displayed correctly.

10. Conclusion

The program calculates apartment-wise electricity bills with progressive tariffs, fixed charges, and rebates. It identifies discrepancies between recorded and calculated bills and ranks apartments by consumption.